

Sales Insights Dashboard Project Report

Online retail sales and profitability analysis using Power BI

Project Overview

This project was built for a hypothetical mid-size U.S. retailer selling Furniture, Office Supplies, and Technology products across four regions. Leadership needed a self-service dashboard that could replace repeated spreadsheet exports and provide a clear weekly view of sales, profitability, seasonality, geography, discounting, and customer segments.

Dataset Summary

The dataset contains 7,649 order-line records from January 2023 through December 2025. It includes order and shipping dates, customers, segments, regions, states, products, quantities, prices, discounts, sales, and profit. The source file contained no null values, used consistent date formats, and was ready for direct validation and modeling in Power BI.

Order Lines	Date Range	Categories	Segments
7,649	Jan 2023-Dec 2025	Furniture, Office Supplies, Technology	Consumer, Corporate, Home Office

Business Questions

Q1: How are sales, profit, margin, order volume, and year-over-year growth performing?

Answer: \$2.06M in sales produced \$449.75K in profit at a 21.9% margin across 8K orders, with 42.3% YoY growth.

Q2: Which months consistently drive the strongest demand?

Answer: November and December create the strongest recurring peaks in all three years, well above the \$49.0K monthly median.

Q3: Which categories and sub-categories generate the most sales and profit?

Answer: Furniture leads with \$1.00M in sales and \$0.24M in profit; Tables are the largest Furniture sub-category at \$0.48M in sales.

Q4: Which regions and states are leading or trailing?

Answer: Central leads at \$0.56M in sales, East trails at \$0.43M, and Missouri is the strongest state in the selected drill-down at \$197K.

Q5: Where may discounting be reducing profitability?

Answer: Furniture contains several high-discount points, while Technology earns strong profit at lower discounts, indicating that discount rules should vary by sub-category.

Q6: How do Consumer, Corporate, and Home Office customers compare?

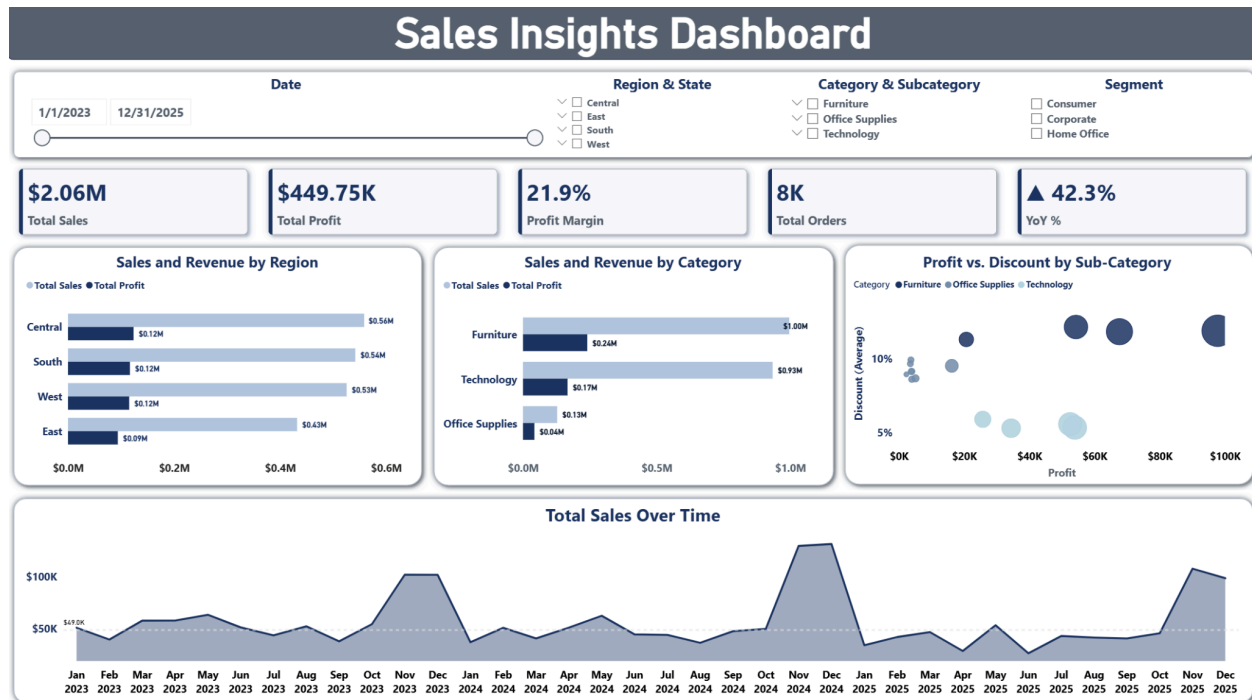
Answer: Consumer provides the most scale, Corporate has the highest margin at 22.3%, and Home Office has the strongest YoY growth at 45.6%.

Key Findings and Recommendations

- Prepare inventory, staffing, and fulfillment capacity before November, using October sales as an early Q4 signal.
- Protect margin with sub-category-specific discount guardrails rather than one universal promotional policy.
- Review East-region pricing, assortment, and local demand drivers because it trails the other regions.
- Use segment-specific tactics: preserve Consumer scale, deepen Corporate relationships, and test targeted Home Office growth campaigns.

Dashboard Development in Power BI

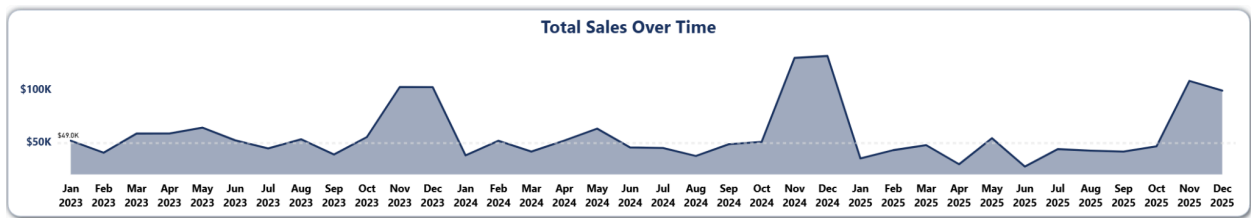
The report was designed as a single-page, self-service dashboard. Slicers set the business context, KPI cards summarize performance, supporting visuals explain the result, and the monthly trend chart highlights recurring patterns.



Development Process

- 1. Data validation:** Loaded the CSV into Power BI, reviewed column types, confirmed the three-year date range, and checked that date, quantity, price, discount, sales, and profit fields were analysis-ready.
- 2. Data model:** Created a dedicated Date table, related it to Order_Date, and marked it as the model's date table so monthly and year-over-year calculations would use a consistent calendar.
- 3. Core measures:** Built reusable DAX measures for Total Sales, Total Profit, Profit Margin %, distinct Order Count, and YoY Growth. Measures were used instead of repeated column aggregations so every visual followed the same definitions.
- 4. Time analysis:** Placed monthly sales on a continuous 2023-2025 timeline and added a median reference line to provide a stable benchmark that is less influenced by the recurring holiday spikes.
- 5. Drill-down design:** Created Region > State and Category > Sub-Category hierarchies, allowing leadership to move from summary results to the underlying geographic and product drivers without leaving the page.
- 6. Visual construction:** Used KPI cards for executive metrics, paired Sales and Profit bars for direct comparison, and a discount-versus-profit scatter plot to surface sub-category patterns that totals alone would hide.
- 7. Interactivity:** Added Date, Region/State, Category/Sub-Category, and Segment slicers, then tested cross-filtering so every KPI and chart updates consistently under the same selection.
- 8. Design and quality assurance:** Applied a restrained slate-and-navy palette, standardized currency units and labels, aligned visual containers, reduced unnecessary color, and tested hierarchy expansion, slicer combinations, and visual readability before capturing the final dashboard.

Sales Trend and Seasonality



Question: How are sales trending over time, and which months consistently drive the strongest demand?

Answer: Monthly sales move above and below the \$49.0K median, but November and December produce clear peaks in 2023, 2024, and 2025. The repeated pattern shows a consistent seasonal effect rather than a one-time spike.

Recommendation: Begin inventory, staffing, and fulfillment preparation before November and use October sales as an early indicator of Q4 demand.

Customer Segment Comparison

The Segment slicer was used instead of a separate segment chart to preserve dashboard space. Selecting each segment updates the same KPI cards, creating a controlled like-for-like comparison.

Consumer

\$1.12M Total Sales	\$241.14K Total Profit	21.6% Profit Margin	4K Total Orders	▲ 42.6% YoY %
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Corporate

\$580.76K Total Sales	\$129.62K Total Profit	22.3% Profit Margin	2K Total Orders	▲ 39.9% YoY %
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Home Office

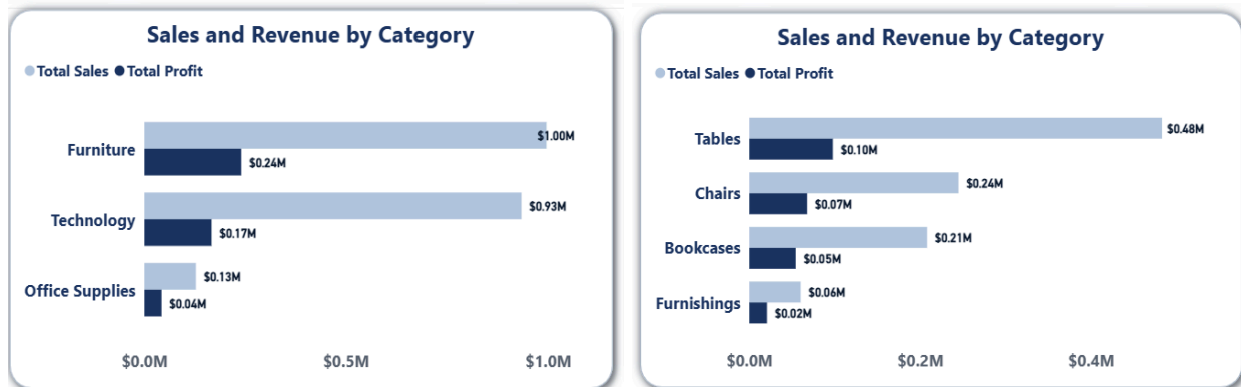
\$361.50K Total Sales	\$78.99K Total Profit	21.8% Profit Margin	1K Total Orders	▲ 45.6% YoY %
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Question: How do the customer segments compare in scale, profitability, and growth?

Answer: Consumer leads in scale with \$1.12M in sales and \$241.14K in profit. Corporate converts revenue most efficiently with a 22.3% margin, while Home Office is the smallest segment but has the strongest YoY growth at 45.6%.

Recommendation: Maintain broad Consumer coverage, prioritize high-value Corporate accounts, and test targeted expansion for the faster-growing Home Office segment.

Category and Sub-Category Performance (Drill-Down)

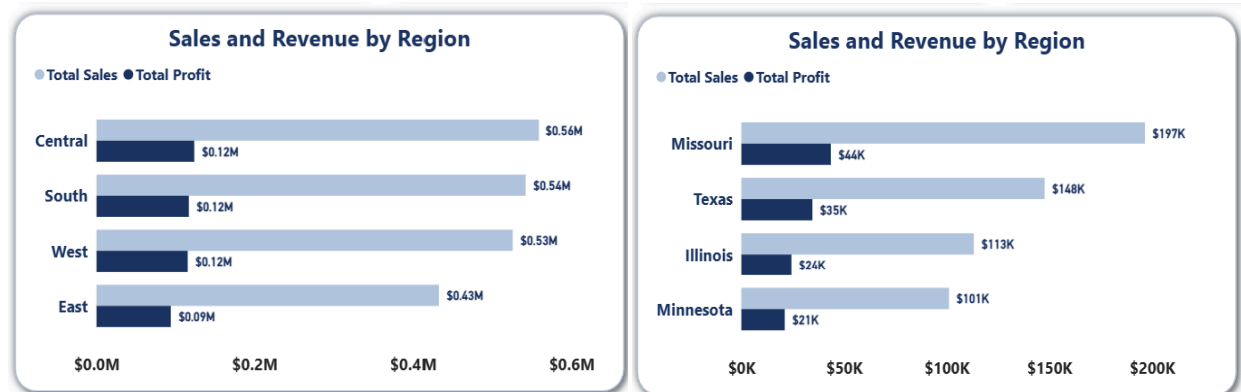


Question: Which categories generate the most sales and profit, and what drives the leading category?

Answer: Furniture leads with \$1.00M in sales and \$0.24M in profit. Within Furniture, Tables are the main driver at \$0.48M in sales and \$0.10M in profit, followed by Chairs and Bookcases.

Recommendation: Protect the high-value Furniture portfolio while reviewing Tables and Chairs individually for margin, discount, and inventory opportunities.

Regional and State Performance (Drill-Down)

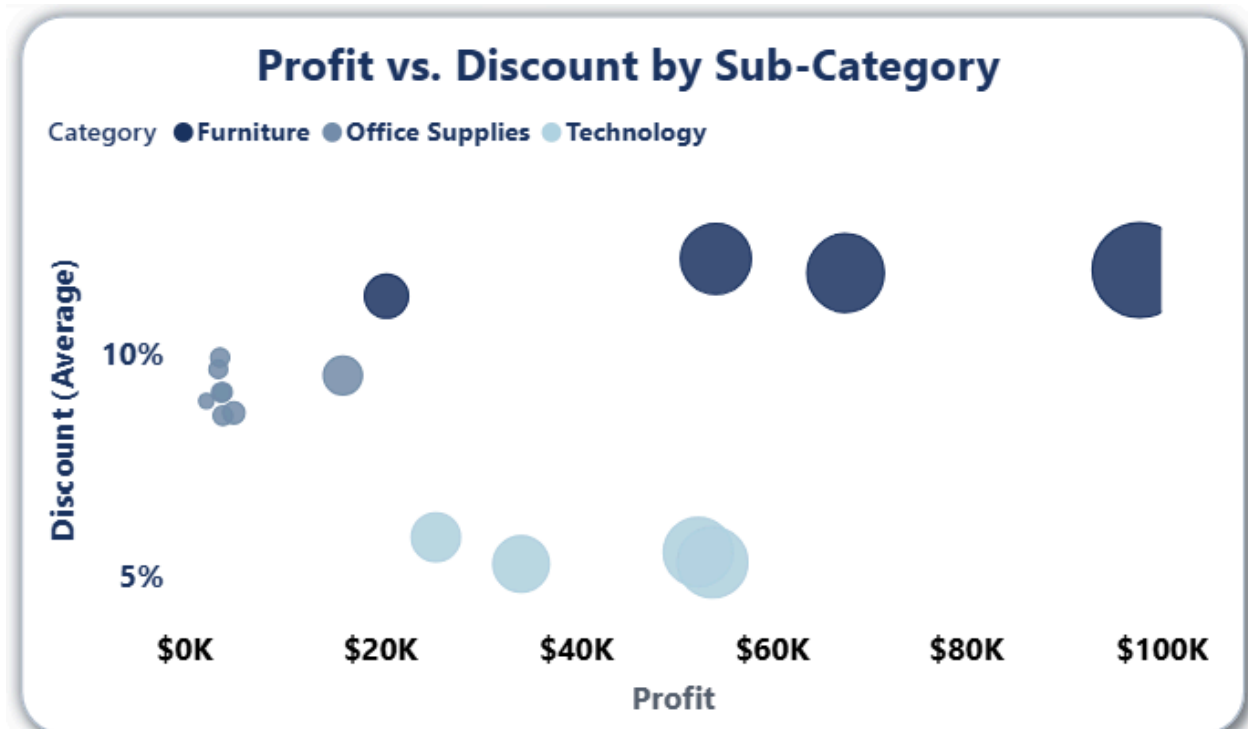


Question: Which regions are leading or trailing, and which states explain the regional result?

Answer: Central ranks first at approximately \$0.56M in sales, while East trails at about \$0.43M. In the state-level drill-down, Missouri leads with \$197K in sales and \$44K in profit, followed by Texas at \$148K and \$35K.

Recommendation: Investigate East-region assortment, pricing, and customer mix, while using high-performing Central states as benchmarks for regional execution.

Discounting and Profitability



Q1: Is discounting helping or hurting profit, and where should management investigate?

Answer: Technology sub-categories produce meaningful profit while generally remaining at lower average discount levels. Furniture contains several of the highest-discount points, showing that profit depends on product economics and sales volume rather than discount rate alone.

Recommendation: Set sub-category-specific discount limits, investigate low-profit items before expanding promotions, and compare margin changes before and after major discount campaigns.

Why This Visual Matters

The scatter plot allows management to evaluate discount and profit simultaneously instead of reviewing either metric in isolation. Category colors preserve context, while each point represents a sub-category that can be investigated further through the dashboard's filters and hierarchy.